

Dana Ferranti

CONTACT INFORMATION

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CURRENT POSITION

Worcester Polytechnic Institute Department of Mathematical Sciences
Assistant Research Professor
Advisor: Sarah Olson

2023 -

RESEARCH INTERESTS

- Computational methods for fluid-structure interaction at low Reynolds number, particularly the method of regularized Stokeslets.
- Biological applications of Stokes flows.
- Reduced ODE models for systems of hydrodynamically coupled bodies.

EDUCATION

Tulane University, New Orleans, LA

2017–2023

PhD, Mathematics

Thesis: *Regularized Stokeslet surfaces and a coupled oscillator system in Stokes flow*

Advisor: Ricardo Cortez

Clark University, Worcester, MA.

2010–2014

BA, Mathematics and Computer Science, Magna Cum Laude

AWARDS

Outstanding Thesis Award (Monetary award)
Tulane University Department of Mathematics

2024

Outstanding Graduate Instructor Award (Monetary award)
Tulane University Department of Mathematics

2023

PUBLICATIONS

- Regularized Stokeslet surfaces. *Journal of Computational Physics*. Volume 508, 113004 (2024). (DF, R. Cortez)
- Generalized matching preclusion in bipartite graphs. *Theory and Applications of Graphs*. Volume 5, Iss. 1, Article 1 (2018). (with Z. Wheeler, E. Cheng, DF, L. Liptak, K. Nataraj)
- The value of prior knowledge in machine learning of complex network systems. *Bioinformatics*. Volume 30, 22 (2017). (DF, D. Krane, D. Craft)

RESEARCH EXPERIENCE

- **Worcester Polytechnic Institute**
Department of Mathematical Sciences

2023 -

- Linear stability analysis of an elastic-structure fluid interaction problem using the method of regularized Stokeslets.
- Mathematical modeling of cellular functions.

- **Tulane University**

2017-2023

Center for Computational Science in Mathematics Department

- Extending the method of regularized stokeslets by using exact integration over triangulated surfaces.
- Reduced models of cilia interaction to investigating the potential effect of elastic coupling and inertia on synchronization.

- **Massachusetts General Hospital**

2016–2017

Physics Research in Department of Radiation Oncology

- Using theoretical models to demonstrate the value of prior knowledge in determining causal relationships in complex networks, with applications to machine learning in medicine.
- Advisor: David Craft.

TEACHING EXPERIENCE	<i>As instructor</i>	
	Worcester Polytechnic Institute	
	• Calculus IV	Spring 2024
	Tulane University	
	• Probability & Statistics I (Math 1110)	Spring 2023
	• Introduction to Applied Math (Math 2240).	Fall 2021
	<i>As teaching assistant</i>	
	Tulane University	
	• Introduction to Applied Math (Math 2240).	2019, 2020, 2021
	• Linear algebra (Math 3090).	2020
	• Calculus I (Math 1210).	2017, 2019
	• Calculus II (Math 1220).	2018, 2020
	• Calculus III (Math 2210).	2018
SERVICE AND OUTREACH	• Reviewer for <i>Pi Mu Epsilon</i> journal	2024
	• Volunteer at WPI's Sonia Kovalevsky Day	2024
	• President of AMS Graduate Student Chapter	2019-2021
	• Mathematics department tea time organizer	2018-2022
	• Treasurer of AMS Graduate Student Chapter	2017-2019
	• Member of Inclusivity in Mathematics Task Force at Tulane (IMTF)	2020-2023
TALKS	• <i>Numerical stability analysis of a coupled fluid-elastic structure system using the method of regularized Stokeslets</i> 77th Annual Meeting of the Division of Fluid Dynamics (November 25, 2024)	
	• <i>Linear stability analysis of a coupled fluid-structure system using the method of regularized Stokeslets</i> Joint annual meeting of Korean Society for Mathematical Biology and Society for Mathematical Biology (July 1, 2024)	
	• <i>Regularized Stokeslet Surfaces</i> Division of Fluid Dynamics (APS Meetings) in Washington D.C. (November 20, 2023)	
	• <i>Simulating bodies immersed in viscous flows: new developments in the Method of Regularized Stokeslets (MRS)</i> Worcester Polytechnic Institute Mathematics Colloquium (September 8, 2023)	
	• <i>Regularized Stokeslet Surfaces</i> Scientific Computing Around Louisiana (March 11, 2023)	
	• <i>Regularized Stokeslet Surfaces</i> Math for All in NOLA (February 25, 2023)	
	• <i>An Extension to the Method of Regularized Stokeslets</i> Special session on Recent Developments in Numerical Methods for PDEs, Joint Math Meetings 2023 (January 4, 2023)	
	• <i>Computational Modeling of Bodies Immersed in Viscous Fluids</i> Hunter College Applied Math Seminar (November 3, 2022)	
CONFERENCES	• Division of Fluid Dynamics (APS Meetings) in Washington, D.C. (November 2023)	
	• Joint Math Meetings in Boston, MA (January 2023)	
	• SIAM Annual Meetings in Pittsburgh, PA (July 2022)	
	• Blackwell-Tapia Conference at IMSI in Chicago, IL (Nov 2021)	
	• Math for All in New Orleans (2020, 2021, 2023)	
	• Scientific Computing Around Louisiana (2018, 2019, 2023)	